

AWS Amplify Cheat Sheet

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Why AWS Amplify?

AWS Amplify is useful for SMEs that want to develop and deploy web or mobile applications without managing complex cloud infrastructure.

It solves the problem of manually configuring and maintaining multiple cloud services such as authentication, storage, APIs and hosting. Amplify automates the creation and deployment of these services, allowing developers to focus on application development.

Example for a Swiss SME:

A company developing a customer portal, an internal business application or a mobile application can use Amplify to quickly deploy cloud services without maintaining its own servers and infrastructure.

Inputs

- Application source code
- Configuration files
- User requests and uploaded data

Outputs

- Hosted web or mobile application
- APIs and backend services
- Stored files and application data
- Monitoring and deployment information

Benefits

- Fast deployment of full-stack applications
- Simplified integration with AWS services
- Built-in CI/CD support
- Scalable infrastructure managed by AWS
- Reduced operational workload
- Suitable for modern frontend frameworks

Limitations

- Strong dependency on the AWS ecosystem
- Less flexibility than manually configured AWS architectures
- Can become complex for highly customized solutions
- Costs may increase with application growth
- Some advanced AWS features require direct AWS configuration

Cost Structure

AWS Amplify uses a pay-as-you-go pricing model.

Costs depend on:

- Build and deployment time
- Hosting bandwidth usage
- Storage consumption
- Backend services used (Cognito, S3, Lambda, AppSync, etc.)
- Build minutes
- Amount of data stored
- Data transfer volume
- Number of requests depending on the AWS services used

Monthly Cost Example

Small SME web application

- 5 developers
- 1 production website
- 20 GB hosting traffic per month
- User authentication with Cognito
- File storage on S3

Estimated monthly cost

- Amplify Hosting: ~10–20 USD
- Storage and authentication: ~5–15 USD
- Total: ~15–35 USD/month

Getting Started

Requirements

- AWS account
- Git repository (GitHub, GitLab, Bitbucket or CodeCommit)
- Node.js installed locally
- AWS credentials configured

Setup

1. Install the Amplify CLI.
2. Configure AWS credentials.
3. Initialize an Amplify project.
4. Add the required services (authentication, API, storage, etc.).
5. Deploy the infrastructure to AWS.
6. Connect the frontend application to the generated backend resources.

Hello World Example

1. Create a simple React application.
2. Initialize Amplify.
3. Add user authentication.
4. Deploy the application.

Common Commands and Operations

Command	Description
<code>amplify configure</code>	Configure AWS credentials
<code>amplify init</code>	Initialize an Amplify project
<code>amplify add auth</code>	Add user authentication
<code>amplify add api</code>	Add a REST or GraphQL API
<code>amplify add storage</code>	Add cloud storage
<code>amplify push</code>	Deploy local changes to AWS
<code>amplify pull</code>	Download an existing Amplify configuration
<code>amplify status</code>	Display pending changes
<code>amplify env list</code>	List environments
<code>amplify env checkout</code>	Switch environment
<code>amplify remove</code>	Remove a resource
<code>amplify delete</code>	Delete the project